

Detailed Project Report (DPR)

"Chittur Kisan Samriddhi" — A ₹10 Crore Catch-Up Investment in Palakkad's Forgotten Drought Belt

Integrated Water, Farm-Productivity & Cooperative-Livelihood Programme

Target region: Eruthenpathy, Vadakarapathy & Kozhinjampara Grama Panchayats, Chittur Block/Taluk, Palakkad District, Kerala (Kerala–Tamil Nadu border)

Implementing partner: Chittur Agriculture Marketing Co-operative Society Ltd.

Prepared for: Submission to IFFCO Foundation / Cooperative Development & CSR Desk, IFFCO — with the water/watershed component proposed for co-funding through the Grameen Vikas Trust (IFFCO & KRIBHCO) and the welfare/insurance component through IFFCO-Tokio CSR.

Endorsed by: [Director, IFFCO — Tamil Nadu / Kerala / Puducherry zone] · [Chittur–Thathamangalam MLA] · [Chairman, Chittur Municipality]

Total outlay sought: ₹10.00 crore (₹1,000 lakh) · **Duration:** 3 years (phased)

Document status: Planning-grade DPR. All figures marked in **[brackets]** are indicative and to be firmed up with (a) a field baseline survey and beneficiary census, (b) engineering estimates for water and civil works, and (c) formal machinery quotations before sanction.

1. Executive summary

The three target panchayats form the **driest, most water-stressed corner of Kerala** — the rain-shadow belt along the Palakkad Gap on the Tamil Nadu border — yet they have received **little dedicated IFFCO development investment**, despite being an intensely agrarian region growing paddy, groundnut, coconut, sugarcane and vegetables with a large small-holder and Tamil-speaking farming community.

This DPR proposes a **₹10 crore, three-year catch-up programme**, delivered through the **Chittur Agriculture Marketing Co-operative Society**, structured around four outcomes:

- Secure water** — recharge, harvesting structures and micro-irrigation (the binding constraint; 35% of outlay).
- Cut input cost and lift yields** — soil testing, balanced nutrition and Nano Urea / Nano DAP.
- Improve realised prices** — cooperative storage, grading, primary processing and market linkage to IFFCO Bazar / iMandi.
- Build a self-sustaining, farmer-owned service institution** — an agri-service hub, custom hiring centre, drone service and FPO owned and operated by the cooperative.

The programme is designed so that the **revenue-generating assets pay for their own operation** after the grant, and so that the ₹10 crore is **leveraged further** through convergence with PMKSY, RKVY, watershed, FPO and state schemes plus NCDC/bank credit.

Headline	Value
Total project cost	₹10.00 crore
Programme duration	3 years (Y1 ₹3.50 cr · Y2 ₹4.00 cr · Y3 ₹2.50 cr)
Direct beneficiaries	~[10,000–12,000] farming households
Anchor spend	Water security & NRM (₹3.50 cr / 35%)
Implementing body	Chittur Agriculture Marketing Co-operative Society Ltd.
Technical partner	KVK Palakkad / Kerala Agricultural University

2. The equity case — why this region, why now

- Chronic neglect.** IFFCO's flagship rural-development, model-village and CORDET-type investments have historically concentrated near its plant and operational hubs in other states, largely **bypassing this border belt**.

- **Highest need, lowest support.** The region faces **Kerala's harshest rainfall deficit and groundwater stress**, heavy borewell dependence, and rising input costs — precisely the problems IFFCO's product-and-fund basket is built to solve — yet it has not been a beneficiary.
- **A ready, trusted delivery vehicle already exists** in the Chittur Agriculture Marketing Co-operative Society, so a grant converts directly into on-ground delivery, not new overheads.
- **Cooperative-to-cooperative equity.** As a farmer-owned cooperative, IFFCO investing in an under-served cooperative-farming community inside its own TN/Kerala/Puducherry zone is squarely within its mandate.

In short: a region that has long powered Palakkad's agriculture has been left off IFFCO's development map. This ₹10 crore corrects that.

3. Situation analysis

3.1 Agro-climatic context

The three panchayats sit in the Palakkad Gap rain-shadow, receiving markedly **lower and more erratic rainfall** than the rest of Kerala. Agriculture depends heavily on **borewells and lift irrigation**, and groundwater tables are under stress from over-extraction. Summer water scarcity is acute and recurrent.

3.2 Cropping and community

- **Principal crops:** paddy, groundnut, coconut, sugarcane, and a strong **vegetable economy**; mango and pulses in pockets.
- **Community:** a large population of small, marginal and tenant farmers, significant women's participation (Kudumbashree), and a sizeable Tamil-speaking farming population reflecting the border location.

3.3 Core problems

1. **Water insecurity** and falling water tables; over-dependence on borewells.
2. **High input costs** and unbalanced fertilisation degrading soil health.
3. **Post-harvest losses** and weak price bargaining power (distress sales at harvest).
4. **Limited access** to modern services — soil testing, machinery, drones, advisory, insurance.
5. **An under-utilised marketing cooperative** operating well below its potential.

3.4 Opportunity

IFFCO's product and service basket, plus its social funds, map almost exactly onto these gaps, and a locally trusted cooperative exists to deliver them. This alignment makes the region an ideal, high-impact, replicable investment.

4. Alignment with IFFCO's mandate

The programme advances IFFCO's core objectives: **strengthening the cooperative movement**, promoting **balanced and sustainable fertilisation (including Nano Urea and Nano DAP)**, improving **farmer incomes and welfare**, and extending **rural development and natural-resource management**. It is therefore a strong fit for IFFCO Foundation / cooperative-development funding, the GVT integrated-development model, and IFFCO-Tokio CSR.

5. Goal and objectives

Goal: Sharply raise the water security, farm income and resilience of ~[10,000–12,000] farming families across the three panchayats through a cooperative-led, IFFCO-funded programme.

Objectives 1. Bring **[1,500] acres** under assured micro-irrigation and build **[540] water recharge/harvesting structures**. 2. Reduce **input cost per acre by [20]%** and improve soil health across **[10,000] farms** via balanced/nano nutrition. 3. Establish a fully equipped **Cooperative Agri-Service Hub** (inputs, soil testing, machinery, drones, advisory, insurance, marketing). 4. Create **storage, grading and primary value-addition** capacity and link produce to IFFCO Bazar / iMandi. 5. Extend **insurance/welfare cover** to **[8,000] families** and scholarships to **[500] students**. 6. Make the cooperative's new service lines **financially self-sustaining by Year 3**.

6. Programme components, specifications and physical targets

Component 1 — Water security & natural-resource management (*anchor, ₹350 lakh*)

- **Micro-irrigation (drip/sprinkler)** on ~[1,500] acres, prioritising vegetables, groundnut and coconut, with IFFCO as top-up over PMKSY subsidy.

- **Recharge structures** — ~[300] borewell/open-well recharge pits and rooftop/field recharge.
- **Farm ponds** — ~[200] lined/unlined ponds for supplemental and lifesaving irrigation.
- **Check dams / water-body revival** — ~[40] structures and desilting/restoration of local ponds and canals.
- Micro-watershed treatment on identified catchments across the three panchayats; user/beneficiary groups formed for maintenance.

Component 2 — Soil health & nano/balanced nutrition (₹50 lakh)

- **Mini soil-testing facility** at the hub plus ~[10,000] soil samples and **soil-health cards** with crop-specific nutrient plans.
- ~[300] **Nano Urea / Nano DAP demonstration plots** on paddy, groundnut, vegetables and coconut, with measured yield and cost data.
- Promotion of **bio-fertilisers, water-soluble and specialty products**, and integrated nutrient management (INM).

Component 3 — Cooperative Agri-Service Hub (₹180 lakh)

- **Hub building** (~[3,000] sq ft): input depot, soil lab, advisory desk, training hall, office and secure storage.
- **IFFCO input depot** fit-out and opening stock — fertilisers, nano products, bio-inputs, seeds, PP chemicals at cooperative rates.
- **Custom Hiring Centre (CHC)** — tractors, power tillers, sprayers, seed drills, groundnut decorticator, paddy thresher and dryers for member rental.
- **Kisan Drone service** — ~[3] agri-drones with batteries/charging and trained local pilots, for nano-fertiliser and PP spraying.
- **Digital advisory** — enrolment on the IFFCO Kisan app plus Malayalam/Tamil voice/SMS advisory with weather and mandi-price alerts.

Component 4 — Market linkage & value addition (₹210 lakh)

- **Godown/storage** (~[2,000] MT) to cut distress sales and post-harvest loss.
- **Grading, cleaning and weighing** unit for quality-based pricing.
- **Primary processing** — groundnut decortication and oil expeller, mini rice unit, cleaning and packaging.
- **FPO** federating farmers across the three panchayats; **branding** and onboarding onto **IFFCO Bazar / iMandi** with buyer tie-ups.

Component 5 — Welfare & risk cover (₹60 lakh)

- **IFFCO-Tokio** crop, cattle, health and accident cover for ~[8,000] families, with premium support and enrolment camps.
- **Kisan Sewa Trust** scholarships for ~[500] farmers' children and a distress-relief fund.

Component 6 — Capacity building (₹60 lakh)

- **Farmer training** for ~[5,000] farmers (CORDET-linked) on water-efficient and balanced-nutrition practices, FPO management and digital tools.
- **Drone-pilot and agri-entrepreneur training** for ~[30] local youth.
- Annual **Kisan Mela** (three editions) with KVK Palakkad / KAU.
- **IEC / awareness** in Malayalam and Tamil, and advisory-app onboarding.

Component 7 — Programme management, M&E and contingency (₹90 lakh)

- **Project Management Unit** (project manager, two field coordinators, MIS/accounts) for three years.
- **Baseline and endline surveys** and third-party evaluation.
- **Audit, documentation and digital dashboard.**
- **Contingency** for cost variation.

7. Detailed budget — ₹10.00 crore (₹1,000 lakh)

#	Component	Line item	₹ lakh	Sub-total
1	Water security & NRM	Micro-irrigation (drip/sprinkler), ~1,500 acres	150	
		Recharge structures (borewell/open-well), ~300	90	
		Farm ponds, ~200	60	
		Check dams / water-body revival, ~40	50	350
2	Soil health & nutrition	Mini soil lab + ~10,000 tests + soil-health cards	20	
		Nano demo plots (~300) + input kits	18	
		Bio-fertiliser / INM promotion & starter inputs	12	50
3	Cooperative Agri-Service Hub	Hub building (~3,000 sq ft)	60	

#	Component	Line item	₹ lakh	Sub-total
		Input depot fit-out + opening stock	20	
		Custom Hiring Centre machinery	70	
		Kisan drones (~3) + training	30	180
4	Market linkage & value addition	Godown / storage (~2,000 MT)	90	
		Grading, cleaning & weighing unit	40	
		Primary processing (groundnut/rice/packaging)	50	
		FPO formation, branding, market linkage	30	210
5	Welfare & insurance	IFFCO-Tokio premium support (~8,000 families)	30	
		Kisan Sewa Trust scholarships (~500)	20	
		Distress-relief fund	10	60
6	Capacity building	Farmer training (~5,000)	22	
		Drone-pilot & agri-entrepreneur youth training (~30)	8	
		Annual Kisan Mela (3 editions)	18	
		IEC / awareness & app onboarding	12	60
7	Programme mgmt, M&E, contingency	Project Management Unit (3 yrs)	45	
		Baseline/endline + third-party evaluation	15	
		Audit, documentation, dashboard	10	
		Contingency	20	90
		TOTAL		1,000

Share by component: Water 35% · Market & value addition 21% · Service Hub 18% · Programme management 9% · Welfare 6% · Capacity building 6% · Soil & nutrition 5%.

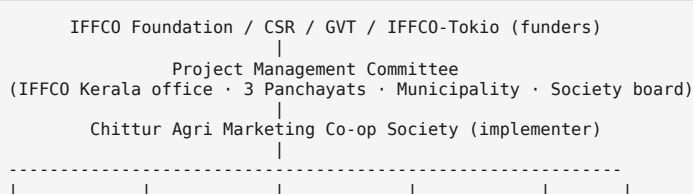
8. Means of finance and leverage

The ₹10 crore IFFCO grant is the anchor. On-ground impact is stretched by **convergence**:

Source	Indicative role
IFFCO Foundation / CSR grant	Core programme grant (₹10 cr sought)
Grameen Vikas Trust (IFFCO & KRIBHCO)	Co-funding of the water/watershed component
IFFCO-Tokio CSR	Insurance awareness and welfare camps
PMKSY ("Per Drop More Crop")	Micro-irrigation subsidy layered under Component 1
RKVY / watershed / soil-health schemes	Demonstrations, soil cards, NRM works
FPO promotion schemes (SFAC/NABARD)	FPO formation and equity grant
NCDC / cooperative banks	Working capital and asset credit for the Society
State department / municipality / panchayats	Land, convergence funds, last-mile mobilisation

9. Implementation model and institutional arrangement

- The **Chittur Agriculture Marketing Co-operative Society** is the single implementing platform and **owns the created assets** (hub, machinery, drones, storage, processing). IFFCO provides the grant, products, technology and training.
- A **Project Management Committee (PMC)** governs delivery, with representation from IFFCO's Kerala State Marketing Office, the three panchayats, the Chittur Municipality and the Society's board.
- KVK Palakkad / Kerala Agricultural University** is the technical partner for agronomy, demonstrations and training.
- A dedicated **Project Management Unit (PMU)** handles day-to-day execution, MIS and accounts.



10. Beneficiary analysis and coverage

Panchayat	Indicative farm households*	Priority focus
Eruthenpathy	[~3,500]	Groundnut, vegetables, borewell recharge
Vadakarapathy	[~4,000]	Paddy, coconut, micro-irrigation
Kozhinjampara	[~4,000]	Vegetables, sugarcane, market linkage
Total (direct)	[~11,500]	

*Placeholders — to be confirmed from panchayat and census data during the baseline survey.

- **Priority inclusion:** small/marginal and tenant farmers, women farmers and Kudumbashree groups, and SC/ST and Tamil-speaking farming families.
- **Indirect beneficiaries:** the wider Chittur block through the strengthened cooperative and a replicable model-village template.

11. Implementation phasing and fund release

Phase	Focus	₹ crore
Year 1 — Foundation	Baseline survey, PMC/PMU set-up, MoU, hub + input depot, soil testing, first nano demos, insurance camps, water planning + first structures	3.50
Year 2 — Build-out	Bulk water/micro-irrigation works, Custom Hiring Centre + drones, godown/grading, FPO formation, market linkage	4.00
Year 3 — Consolidation	Primary processing unit, scale-up of proven practices, financial self-sufficiency of services, documentation and hand-over	2.50
	Total	10.00

Phases are project stages, not fixed calendar promises; sequencing of water works follows the rainfall calendar and approvals.

12. Sustainability and financial viability of service lines

The hub is designed to be **self-financing after the grant** through revenue-generating services owned by the Society. Indicative steady-state (from Year 3) annual figures:

Service line	Basis	Indicative annual revenue (₹ lakh)
IFFCO input depot	Margin on fertiliser/nano/input sales	[18-25]
Custom Hiring Centre	Machinery rental (tractor/thresher/dryer hours)	[10-15]
Drone spraying service	Per-acre service fee	[6-10]
Storage / godown	Storage & handling charges	[5-8]
Marketing commission / value addition	Commission + processing margin	[8-12]
Total		[47-70]

These revenues are expected to comfortably cover operations, maintenance and staffing of the service arm, with surplus reinvested and shared with members — the core reason the programme endures beyond IFFCO's grant. (Figures are illustrative and to be validated by a business plan during Year 1.)

13. Monitoring, evaluation and KPIs

- **Baseline → endline** survey of income, input cost, yield, water table and insurance coverage.
- **Tracked KPIs:** acres under micro-irrigation; recharge structures built; % input-cost reduction; nano-adoption rate; drone-service acres; produce aggregated and average price gain; storage throughput; families insured; scholarships awarded; and **cooperative turnover and surplus**.
- **Review cadence:** quarterly PMC review, annual independent/IFFCO review, and a live digital dashboard maintained by the Society.

14. Governance, transparency and safeguards

- **Ring-fenced, independently audited** project account; public display of works and beneficiary lists.
- **Conflict-of-interest protocol:** office-bearers holding political or municipal posts **recuse themselves** from procurement and beneficiary-selection decisions; procurement follows cooperative rules and competitive quotations.
- **Transparent, needs-based beneficiary selection** prioritising small and marginal farmers.
- **Precondition:** the Society's accounts and audit must be in good standing before disbursement.

15. Risk analysis and mitigation

Risk	Likelihood	Mitigation
Rainfall failure delaying water works	Medium	Sequence works to monsoon; prioritise recharge + micro-irrigation
Cooperative governance/audit gaps	Medium	Pre-sanction audit; PMU oversight; ring-fenced account
Under-utilised machinery/drone	Medium	Demand-linked rollout; service pricing; trained local operators
Slow market-linkage uptake	Medium	Anchor on own aggregation + IFFCO Bazar; buyer MoUs
Convergence funds not materialising	Medium	Grant designed to stand alone; convergence is upside, not dependency
Conflict-of-interest perception	Low-Medium	Formal recusal protocol; transparent procurement

16. Expected outcomes and impact

- A measurable **rise in net farm income** and a **[20]% cut in input cost** for participating households.
- **Restored water security** in a chronically drought-prone belt through recharge and micro-irrigation.
- A **self-sustaining, professionally run cooperative** serving as IFFCO's flagship model rural-service institution in Kerala.
- **Lower post-harvest loss, better prices, wider insurance cover**, and a replicable **model-village template** for the rest of Chittur block.

17. Next steps

1. Endorsement and forwarding by the [IFFCO director] to the IFFCO Foundation / board.
2. **Baseline survey** and detailed engineering estimates to firm up bracketed figures and beneficiary counts.
3. **MoU** between IFFCO (Kerala office), the Society, the three panchayats and KVK Palakkad.
4. Sanction of ₹10 crore and Year-1 launch.

Annexure A — Indicative unit costs (to be validated)

Item	Unit	Indicative unit cost
Micro-irrigation (IFFCO top-up over PMKSY)	per acre	₹[10,000]
Borewell/open-well recharge structure	per unit	₹[30,000]
Farm pond	per unit	₹[30,000]
Check dam / water-body revival	per unit	₹[1.25 lakh]
Soil test + soil-health card	per sample	₹[150-200]
Nano demo plot kit	per plot	₹[5,000-6,000]
Agri-drone (with battery/charging)	per unit	₹[9-10 lakh]

Annexure B — Documents required at appraisal

- Registration certificate and latest audited accounts of the Chittur Agriculture Marketing Co-operative Society.
- Board resolution authorising the project and the IFFCO application.
- Land/space documents for the hub, godown and processing unit.
- Panchayat/municipality endorsement letters and convergence commitments.
- Baseline survey report and detailed engineering/cost estimates.
- Feedstock/market/buyer MoUs and the service-arm business plan.

